InChI Versioning

Based on Discussions at InChI Technical Meeting, Aachen, July 2026

Presented by Ian Bruno at the InChI Meeting in Chicago on 21-Aug-2026

Versioning

There are two entities we version:

- The InChI generator (software)
- The InChI itself (string and key)

Current software version is 1.07.5

- We will be moving to use YEAR.N e.g. 2027.2
- Will use modifiers for test releases e.g. 2027.2-beta1

The following discussion relates to versioning of the InChI itself

Versioning

When InChI for Molecular Inorganics is released, the InChI Generator will generate a different standard InChI **for this class of compounds**

Current consensus is that this should not **be a silent change - there should be something explicit that indicates this is a** new standard for molecular inorganics

There is also a strong a consensus that standard InChIs for compounds not impacted by this change must not change

Discussion across Technical meetings at Aachen have led to the following proposal that we would like to road test here…

Current Version Indicators

Standard InChI:

InChI=**1S**/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3

Standard InChIKey:

RYYVLZVUVIJVGH-UHFFFAOY**SA**-N

Versioning - Proposal

It is proposed to extend the current "1S" designator to indicate when the standard treatment has changed for an input structure. This will only apply to structures impacted by the change.

Imagine a sequence of changes, each in a separate release:

- Molecular Inorganics then Feature X

Versioning would be as follows:

- Organic without feature X
- Molecular inorganic without feature X
- Any molecule with feature X

InChI

1S (as currently)

1SB (will come back to why B)

1SC (regardless of whether molecular inorganic)

Versioning - Proposal

It is proposed to extend the current "1S" designator to indicate when the standard treatment has changed for an input structure. This will only apply to structures impacted by the change.

Imagine a sequence of changes, each in a separate release:

- Molecular Inorganics then Feature X then Feature Y

Versioning would be as follows:

- Organic without feature X or Y
- Molecular inorganic without feature X or Y
- Any molecule with feature X but not Y
- Any molecule with feature Y
  inorganic or has Feature X)

InChI	InChIKey
1S	SA (as currently)
1SB	SB (InChIKey is why we propose B)
1SC	SC
1SD	SD regardless of whether molecular

Proposal Restated / Elaborated

InChI version for a structure should be incremented if a change will mean that the InChIKey generated from its input file will be different as a result of that change

This is because it is not possible to distinguish InChI vintage from a Key in the same way that one can from looking at InChI layers

This probably means that any change to the content of the InChI string (including additional layers or sublayers) should result in an increment in version number

Version indicator could be correlated to a set of changes in a specific release (would lose discrimination)

If a change will impact > 20% of molecules for which InChIs are believed to have previously been generated, that should be considered to be a breaking change to Version 1 (threshold probably needs further discussion)